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Original Research

The Impact of a Robust Supplemental Screening Program on False-Negative Rates: Time for New Benchmarks?

Ashley Huppe, MD,[1,*] Onalisa Winblad, MD,[1] Jessica Peterson, MD,[1] Camron Smith, DO,[1] Molly Hill, MD,[1] Lauren Clark, MS,[1] Carissa Walter, MPH,[1] Allison Aripoli, MD[1]

[1]Department of Radiology, University of Kansas Health System, Kansas City, KS, USA

[*]Corresponding author: Ashley Huppe (ahuppe@kumc.edu)

Abstract

Objective: Screening mammography false-negative rates (FNRs) measure interval cancers (ICs) that are confounded by asymptomatic false-negative (FN) cancers detected on supplemental screening (SS) examinations. The purpose of this study was to 1) evaluate the impact of SS on FNRs for patients at higher–than–average risk in a robust SS program and 2) compare patient and tumor characteristics of symptomatic ICs and asymptomatic FN cancers.

Methods: Screening mammogram audit metrics from an academic institution from July 1, 2018, to June 30, 2023, were retrospectively reviewed. A negative screening mammogram with a subsequent diagnosis of cancer within 12 months was considered an FN cancer. Patient risk, method of detection, SS utilization, and tumor characteristics were collected. Cochran-Mantel-Haenszel tests were used to analyze trends, and univariable tests were used to compare symptomatic and asymptomatic cases.

Results: Of 106 750 screening mammograms, 624 screening-detected cancers and 119 FN cancers were identified, with a cancer detection rate of 5.8/1000 and an FNR of 1.11/1000. There were 53 (45%) symptomatic ICs and 66 (55%) asymptomatic FNs, with a symptomatic FNR of 0.50/1000. Patients with symptomatic IC were more likely to have a personal history of breast cancer ($P = .0289$), and 74% (39/53) were higher-than-average risk. Symptomatic ICs were larger at diagnosis (mean size 2.3 cm vs 1.5 cm; $P = .0293$). MRI detected the majority (37/66, 60%) of asymptomatic FNs.

Conclusion: A robust SS program allows for the symptomatic FNR to fall below published national performance benchmarks from the Breast Cancer Surveillance Consortium. Most patients with symptomatic IC were at higher-than-average risk and underutilized SS MRI.

Key words: breast imaging; screening; false negative; supplemental screening; benchmarks.

Introduction

According to the U.S. Food and Drug Administration (FDA) and the Mammography Quality Standards Act (MQSA), mammography facilities must review false negatives (FN) and interval cancers (ICs) that become known to the facility.[1] An FN is defined as a diagnosis of breast cancer within 12 months of a negative screening mammogram. Symptomatic FN cancers are considered ICs and are known to have worse outcomes.[2-4] Asymptomatic FN cancers are those detected by modalities other than mammography or identified incidentally after surgery.[5,6] The benchmark FN rate (FNR) for screening mammography is an important audit metric to be reviewed. The benchmark FNR of screening mammography is 0.8 per 1000 patients, originally based on Breast Cancer

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Key Messages

- A robust supplemental screening (SS) program for patients at higher-than-average risk allows for symptomatic false-negative rates (FNRs) to fall below historic benchmarks of 0.8 per 1000.
- The purpose of SS is to increase cancer detection, which may result in a higher screening mammography FNRs. Multi-institutional analysis of the asymptomatic FNR in the setting of SS is needed to assess whether new benchmarks may be appropriate.
- Increased utilization of SS MRI may result in further elevation in screening mammography FNRs because 97% (28/29) of patients in our study with symptomatic interval cancers met criteria for but were not utilizing SS MRI based on 2023 American College of Radiology guidelines.

Surveillance Consortium (BCSC) data collected up to 2013, including both symptomatic ICs and asymptomatic FN cancers, and updated to include digital breast tomosynthesis (DBT) in 2023.[7,8]

For patients at higher–than–average risk, supplemental screening (SS) is recommended. Commonly performed SS options include both handheld US (HHUS) and automated US (AUS), full-protocol (FP) MRI, abbreviated (AB) MRI, contrast-enhanced mammography (CEM) and molecular breast imaging. Utilization of SS is expected to increase after the March 10, 2023, FDA final rule amending regulations under MQSA for a national dense breast reporting standard.[9-13]

The American College of Radiology (ACR) outlines clinical practice guidelines for screening higher-than-average-risk patients utilizing robust strength-of-evidence analysis and extensive literature review. The first, published in 2018, upheld the traditional high-risk screening criteria and also included a new recommendation for screening MRI in patients with a personal history (PH) of breast cancer diagnosed before age 50 or with a PH and dense breast tissue.[14] The 2023 updated guidelines include a recommendation for screening MRI in patients with dense breast tissue as their only risk factor.[11] A subsequent update to the ACR Appropriateness Criteria in 2024 outlines the appropriateness of SS examinations for women based on risk and density.[15]

Increasing utilization of SS for patients at higher–than–average risk allows for detection of asymptomatic FN cancers. Current screening mammography benchmarks do not address the potential impact of SS on FNRs. In fact, few studies distinguish between symptomatic ICs and asymptomatic FN cancers.[5,6] The proposed ACR Method of Detection (MOD) is designed to capture the first imaging test or symptom to trigger the workup and diagnosis of breast cancer, characterizing symptomatic from asymptomatic cases at the patient level.[16] Because a reduction in ICs can be considered a surrogate measure for longer-term screening outcomes, such as morbidity and mortality, standard collection of MOD data points should provide information about the impact of screening mammography. An understanding of how SS affects screening mammography benchmarks is also important.[6]

Given the updated screening guidelines, federal breast density notification regulations, and increased utilization of SS, it is expected that the detection of asymptomatic FN cancers will increase the screening mammography FNR for facilities performing significant volumes of SS. However, no established benchmarks or guidelines for FNRs in this setting currently exist.[7] The purpose of this study is to evaluate the impact of SS on FNRs for patients at higher–than–average risk in the setting of a robust SS program. Secondary objectives are to compare patient and tumor characteristics of symptomatic ICs and asymptomatic FN cancers, including analysis of SS utilization in our patient population at higher-than-average risk.

Methods

Selection of study population

This retrospective study at a single academic, National Cancer Institute–designated Comprehensive Cancer Center was approved by the institutional review board with waiver of patient consent. The study was compliant with HIPAA. All patients undergoing screening mammography from July 1, 2018, to June 30, 2023, who had a normal or benign screening mammogram (Breast Imaging Reporting and Data System [BI-RADS] 1 or 2) and a subsequent diagnosis of cancer within 12 months of the mammogram were included and recorded as an FN in the MQSA audit.[17] No cases were excluded. There were no cases of patients who returned early for a subsequent screening mammogram (within the 12-month period) and then diagnosed with breast cancer included in this cohort. All mammograms performed during the study period utilized DBT with synthetic 2D mammography. One-year imaging follow-up was available for all included patients, and all screening mammography is linked to the National Cancer Registry.

Collection of patient demographics and risk factors

Manual review of the mammography reporting system and electronic medical record was performed to collect patient demographic data, breast density, and risk factors for breast cancer known at the time of the FN screening examination. All data were collected and stored using REDCap.[18] Factors contributing to higher-than-average risk status were obtained from the patient-completed mammography screening intake form available to the radiologist in the mammography reporting system at the time of image interpretation. Subcategories for elevated risk included the following: cumulative dose of ≥10 Gy chest radiation before age 30, actionable genetic mutation or untested first-degree relative, PH of breast cancer

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before age 50, PH of breast cancer at any age with dense breast tissue, PH of breast cancer, PH of atypia (including atypical ductal hyperplasia, atypical lobular hyperplasia, and lobular carcinoma in situ), dense tissue (defined as BI-RADS category C or D by the interpreting radiologist), or lifetime risk ≥20% as calculated by a mammography reporting system utilizing the eighth edition of Tyrer-Cuzick (TC).

Collection of imaging characteristics

Imaging records of patients with an FN mammogram were retrospectively reviewed to assess utilization of SS MRI according to the 2018 ACR guidelines (available at the beginning of the study period) and according to the 2023 ACR guidelines (published after the study period).[11,14] Patients with traditional high-risk factors (≥ 20% lifetime risk, actionable genetic mutation or untested first-degree relative, and cumulative dose of ≥10 Gy chest radiation before age 30) as well as those with a PH before age 50 or a PH and dense breast tissue were assessed according to the 2018 recommendations. Given the updated 2023 ACR guidelines for SS MRI, patients with the above risk factors plus those with dense breast tissue only were assessed according to the updated 2023 recommendations.

The characteristics of FN cancers were collected, including the biopsy modality, histology, visibility on mammography according to the radiologist at the time of interpretation as documented in the imaging report, and the dominant imaging finding as described by the interpreting radiologist (utilizing fifth edition BI-RADS terminology).[17] Cases were not reinterpreted to assess retrospective mammographic visibility.

Method of detection for all FN cancers was retrospectively assigned utilizing ACR breast MOD categories.[16] Patient or provider-detected cancers were designated as "symptomatic IC," and the presenting symptom(s) were recorded. False-negative cancers detected on another screening breast imaging examination, incidentally on a non–breast imaging modality or incidentally during surgery were designated as "asymptomatic FN cancers" and characterized accordingly. The imaging categories used in this study align with ACR MOD and include screening breast US (subcategorized for our study into HHUS and AUS), screening breast MRI (subcategorized for our study into FP MRI and AB MRI), and incidental detection by a non–breast imaging examination. At our institution, SS can be performed concurrently with screening mammography or at any time during the 1-year interval after the screening mammogram. Therefore, FN cancers include those detected by SS on the same day after a negative screening mammogram or at any point during the following 1-year period as well as those that presented symptomatically within the subsequent 1-year period.

Collection of institutional audit data and imaging volumes

Volumes for each screening modality by year were recorded, and the FNR was trended over the 5-year study period. Mammography Quality Standards Act audit statistics for the institution were also recorded, including sensitivity, specificity, cancer detection rate (CDR), abnormal interpretation rate, and positive predictive values. Audit metrics were compared with nationally recognized benchmarks published by the BCSC.[7,8]

Statistical analysis

Cochran-Mantel-Haenszel tests were used to analyze trends in the FNR and utilization of SS over the study period. Demographics, patient risks, and diagnostic characteristics between symptomatic and asymptomatic were compared using Student's *t* tests with Satterwaithe degrees of freedom for continuous variables and chi-square or Fisher exact tests for categorical variables. All analyses were performed using SAS 9.4.

Results

Overall FNR and MQSA statistics

A total of 106 750 screening mammograms were performed, with 624 cancers detected. The overall FNR was 1.11 per 1000 (95% CI, 0.91-1.31). The study population included 119 FN mammograms. Symptomatic ICs comprised 45% (53/119; 95% CI, 35.6-53.5). Asymptomatic FN cancers comprised 55% (66/119; 95% CI, 46.5-64.4). The symptomatic FNR was 0.50 per 1000 (95% CI, 0.36-0.63), and the asymptomatic FNR was 0.62 per 1000 (95% CI, 0.47-0.77). The symptomatic FNR trended downward, and the asymptomatic FNR trended upward during the 5-year period, although this did not reach statistical significance (Figure 1). Overall institutional screening mammography audit data are shown in Table 1, with an overall CDR of 5.8 per 1000 during the study period. While the institutional sensitivity of 82.9% was lower than the benchmark of 86.9%, all other audit data met or exceeded the benchmark criteria.

Supplemental screening

There was a statistically significant increase in all SS modalities relative to screening mammography during the study period (Figure 2). Utilization of AB MRI increased the most over the 5-year period, from 108 examinations in year 1 (2018 to 2019, the first year it was offered at our institution) to 966 examinations in year 5 (2022 to 2023). Although CEM is performed at our institution, it was implemented near the end of the study period and did not contribute to the detection of any FN cancers; therefore, it was not included in this subanalysis.

Demographics and patient risk

The mean age for patients with an FN screening mammogram during the study period was 60.0 years (SD 12.4), similar to the mean age of 59.0 years for all patients who underwent screening mammography during the time frame (Table 2). The age at FN was significantly younger for asymptomatic FN at 58.0 years (95% CI, 54.9-61.2), compared with 62.5 years (95% CI, 59.3-65.6) for symptomatic IC ($P = .0487$).

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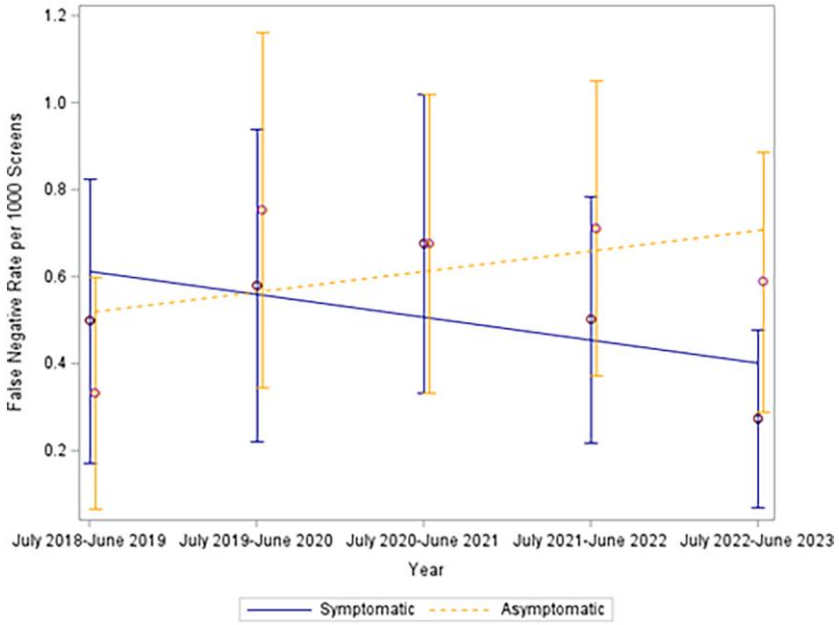


Figure 1. Symptomatic and asymptomatic false-negative (FN) screening examinations per 1000 screening mammograms by year. While the symptomatic FN rate trended down and the asymptomatic rate trended up, these trends did not reach statistical significance ($P = .2151$).

Table 1. Institutional Screening Mammography Audit Data for the Study Period Compared With Nationally Recognized Benchmarks Set Forth by the BCSC[7,8]

Performance metric	Institutional audit data	Benchmarks, BCSC, Lehman et al[7]	Benchmarks, DBT, Lee et al[8]
Cancer detection rate	5.8 per 1000	5.1 per 1000	5.8 per 1000
Abnormal interpretation rate	5.8%	11.6%	8.3%
Sensitivity	82.9%	86.9%	87.4%
Specificity	94.7%	88.9%	92.2%
PPV1	10.0%	4.4%	6.9%
PPV 2	37.5%	25.6%	32.2%
PPV3	38.1%	28.6%	35.5%

Abbreviations: BCSC, Breast Cancer Surveillance Consortium; DBT, digital breast tomosynthesis; PPV1, probability of cancer among those screened; PPV2, probability of cancer among those in whom biopsy was recommended; PPV3, probability of cancer among those who underwent biopsy.

Symptomatic ICs were more common in women with a PH and those with nondense breast tissue. More patients with asymptomatic FNs had a TC score ≥20% at the time of their FN screening mammogram (51.5% [95% CI, 39.5-63.6] vs 26.4% [95% CI, 14.6-38.3]; $P = .0143$), which is typical of a population undergoing SS. There were more minority patients with symptomatic ICs (12/53, 23%) than asymptomatic FNs (5/66, 8%), although this did not reach statistical significance ($P = .0748$). Patient demographics are summarized in Table 2.

Of all FN cancers, 87% (104/119) were retrospectively identified as having at least 1 factor that elevated individual lifetime risk (Table 3). Nearly all, 98% (65/66, [95% CI, 95.5-100]) asymptomatic FNs were categorized as higher-than-average risk, and 74% (39/53 [95% CI, 61.7-85.5]) of symptomatic ICs were higher-than-average risk ($P < .0001$). While there was no difference in the number of asymptomatic FNs and symptomatic ICs in patients with a PH of cancer before age 50, patients diagnosed after age 50 with dense tissue and without dense tissue showed a statistically significant higher number of symptomatic ICs ($P = .0442$ and $P = .0017$, respectively).

Symptomatic IC

There were 33 patients comprising 62% (33/53) of symptomatic ICs with no PH of breast cancer, 39% (13/33) of which had a TC score ≥20%. Presentation of symptomatic ICs is outlined in Figure 3. Most, 87% (46/53) of patients with symptomatic IC, had no SS within the prior year; 86% (6/7) of those who did utilized US screening. When reviewing patient utilization of SS MRI according to ACR guidelines, we found that 45% (24/53) of patients with symptomatic IC fit

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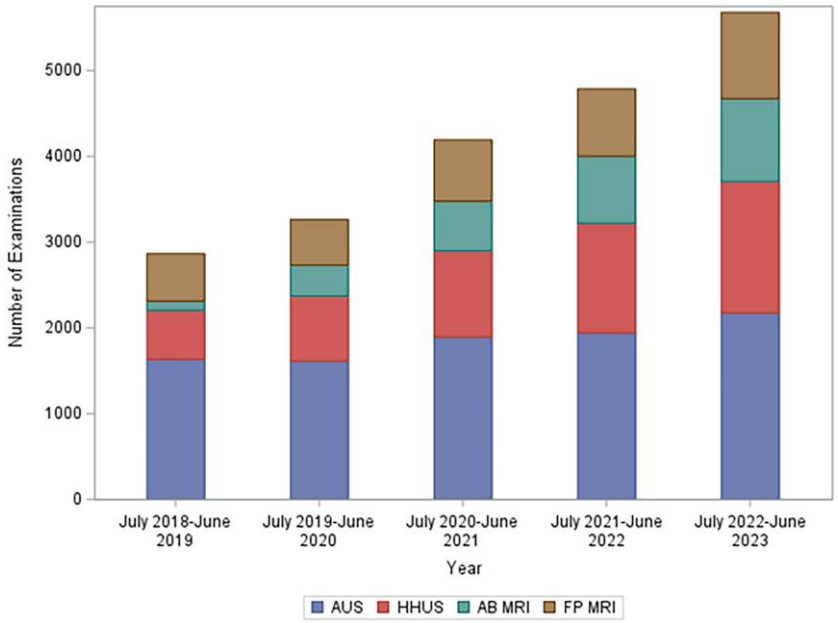


Figure 2. Utilization of supplemental screening modalities over the 5-year study period. Abbreviations: AB MRI, abbreviated MRI; AUS, automated US; FP MRI, full-protocol MRI; HHUS, handheld US.

criteria for SS MRI according to 2018 guidelines, and 55% (29/53) fit the criteria according to the 2023 guidelines (Table 4). Of those who met criteria, 97% (28/29) of patients with symptomatic IC were not utilizing SS MRI according to 2023 guidelines (Figure 4).

Asymptomatic IC

Method of detection for asymptomatic FN is displayed in Figure 5. Of those detected by SS, MRI detected 60% (37/62), followed by 40% (25/62) with US. The mean time to diagnosis from the day of FN mammogram was 147 days, and 18% (11/62) had their FN mammogram concurrent with SS. There were 53% (35/66) of asymptomatic FN patients who were utilizing SS MRI according to both 2018 and 2023 guidelines (Figure 6). No patients with dense tissue as their only risk factor were undergoing SS MRI.

Comparison of symptomatic vs asymptomatic IC

Symptomatic ICs were larger at diagnosis (mean size 2.3 cm [95% CI, 1.8-2.8) vs 1.5 cm [95% CI, 10-2.0]; $P = .0293$) and detected later (mean 226 days [95% CI, 199-252] vs 147 days [95% CI, 124-171] after negative screening mammogram; $P < .001$). Symptomatic ICs were more likely to have morphologically abnormal lymph nodes at the time of diagnostic imaging (36% [95% CI, 22.9-48.8] vs 11% [95% CI, 3.2-18.0]; $P = .0007$]. Just more than one-third of symptomatic ICs, 34% (18/53 [95% CI, 21.2-46.7]) remained mammographically occult at the time of diagnosis. Asymptomatic FNs were more likely to be grade 1 than symptomatic ICs (19% [95% CI, 8.9-27.5] vs 6% [95% CI, 0-11.9]; $P = .0492$). There were no significant differences in histologic subtype, hormone receptor, or HER2 status between symptomatic ICs and asymptomatic FNs. Table 2 provides additional data on the imaging presentation of both symptomatic ICs and asymptomatic FNs.

Discussion

While MQSA regulations require mammography facilities to review FN cancers, the anticipated effects of SS on the screening mammography medical audit are largely unknown because traditional FN audits do not distinguish between symptomatic ICs and asymptomatic FNs. Our study demonstrated a positive impact of SS on the FNR for symptomatic IC, with a symptomatic FNR of 0.50 per 1000, well below the benchmark of 0.8 per 1000. Most patients with symptomatic IC were at higher-than-average risk for breast cancer, with significantly more symptomatic patients with IC having a PH of breast cancer than those with asymptomatic FN cancers. A total of 97% (28/29) of patients with symptomatic IC were underutilizing SS MRI, based on 2023 ACR recommendations.

The overall FNR for our study was 1.1 per 1000. This is higher than the historic benchmark of 0.8 per 1000; however, more than half of the FNs were SS–detected, asymptomatic FNs. The results of this study suggest there is an opportunity to refine FNR benchmarks, potentially with separate benchmarks for symptomatic ICs and asymptomatic FNs. Recent North American studies have reported an FNR of 0.5 per 1000 DBT examinations, 0.6 per 1000 DBT examinations, and 0.6 to 0.9 per 1000 DBT examinations, without reporting the contribution of asymptomatic FNs.[19,20,21] Other studies that subcategorize by symptomatic and asymptomatic FNs demonstrate similar decreases in the symptomatic FNR by varying levels, from 0.6 to 0.4 per 1000 examinations in the

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Table 2. Patient Demographics and Risk Status at the Time of the FN Screening Mammogram

	Symptomatic	Asymptomatic	All	*P*-Value
Overall *N*	53	66	119	-
Age at FN screening mammogram, Mean (SD)	62.5 (11.4)	58.0 (12.8)	60.0 (12.4)	.0487[a]
Ethnicity, *N* (% of column)	-	-	-	.6547[b]
Non-Hispanic/Latino	50 (94.3%)	64 (97.0%)	114 (95.8%)	-
Hispanic/Latino	3 (5.7%)	2 (3.0%)	5 (4.2%)	-
Race, *N* (% of column)	-	-	-	.0913[b]
American Indian/Alaskan Native	1 (1.9%)	0 (0.0%)	1 (0.8%)	-
Asian/Pacific Islander	3 (5.7%)	1 (1.5%)	4 (3.4%)	-
Black	8 (15.1%)	4 (6.1%)	12 (10.1%)	-
White	41 (77.4%)	59 (89.4%)	100 (84.0%)	-
Other	0 (0.0%)	2 (3.0%)	2 (1.7%)	-
History of breast malignancy, *N* (% of column)	-	-	-	.0289[c]
No	33 (62.3%)	53 (80.3%)	86 (72.3%)	-
Yes	20 (37.7%)	13 (19.7%)	33 (27.7%)	-
Breast density, *N* (% of column)	-	-	-	.0110[c]
Not dense (A, B)	30 (56.6%)	22 (33.3%)	52 (43.7%)	-
Dense (C, D)	23 (43.4%)	44 (66.7%)	66 (56.3%)	-
Higher-than-average risk status, *N* (% of column)	-	-	-	<.0001[c]
No	14 (26.4%)	1 (1.5%)	15 (12.6%)	-
Yes	39 (73.6%)	65 (98.5%)	104 (87.4%)	-
Tyrer-Cuzick score in reporting system at time of FN mammogram >20%, *N* (% of column)	-	-	-	.0143[c]
No	19 (35.8%)	19 (28.8%)	38 (31.9%)	-
Yes	14 (26.4%)	34 (51.5%)	48 (40.3%)	-
N/A (due to personal history of breast cancer)	20 (37.7%)	13 (19.7%)	33 (27.7%)	-
Presentation of FN cancer				
Mass	38 (71.7%)	57 (86.4%)	95 (79.8%)	.0475[c]
Calcifications	4 (7.5%)	0 (0.0%)	4 (3.4%)	.0369[b]
Developing asymmetry	3 (5.7%)	0 (0.0%)	3 (2.5%)	.0856[b]
Architectural distortion	2 (3.8%)	0 (0.0%)	2 (1.7%)	.1963[b]
Other	4 (7.5%)	9 (13.6%)	13 (10.9%)	.2899[c]

Abbreviation: FN, false-negative; N/A, not applicable.
[a]Student's *t* test with Satterthwaite degrees of freedom.
[b]Fischer's exact test.
[c]Chi-square test.

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Table 3. Patients With at Least 1 Factor That Elevated Risk

	Symptomatic	Asymptomatic	All	*P*-value
Overall *N*	39	65	104	
List of risks, *N* (% of column)				
>20% based on risk calculation/family history	13 (33.3%)	28 (43.1%)	41 (39.4%)	.3249[a]
Genetic mutation/untested first-degree relative	**1 (2.6%)**	**10 (15.4%)**	**11 (10.6%)**	.0396[a]
PH cancer <50	6 (15.4%)	5 (7.7%)	11 (10.6%)	.2169[a]
PH cancer ≥50 + dense tissue	**4 (10.3%)**	**1 (1.5%)**	**5 (4.8%)**	.0442[a]
PH cancer ≥50	**10 (25.6%)**	**3 (4.6%)**	**13 (12.5%)**	.0017[a]
PH atypia/LCIS	0 (0.0%)	1 (1.5%)	1 (1.0%)	.4364[a]
Dense tissue only	5 (12.8%)	17 (26.2%)	22 (21.2%)	.1070[a]

n = 104/119, 87%; Risk categories that resulted in statistically significant differences are highlighted in bold.
Abbreviations: LCIS, lobular carcinoma in situ; PH, personal history.
[a]Chi-square test.

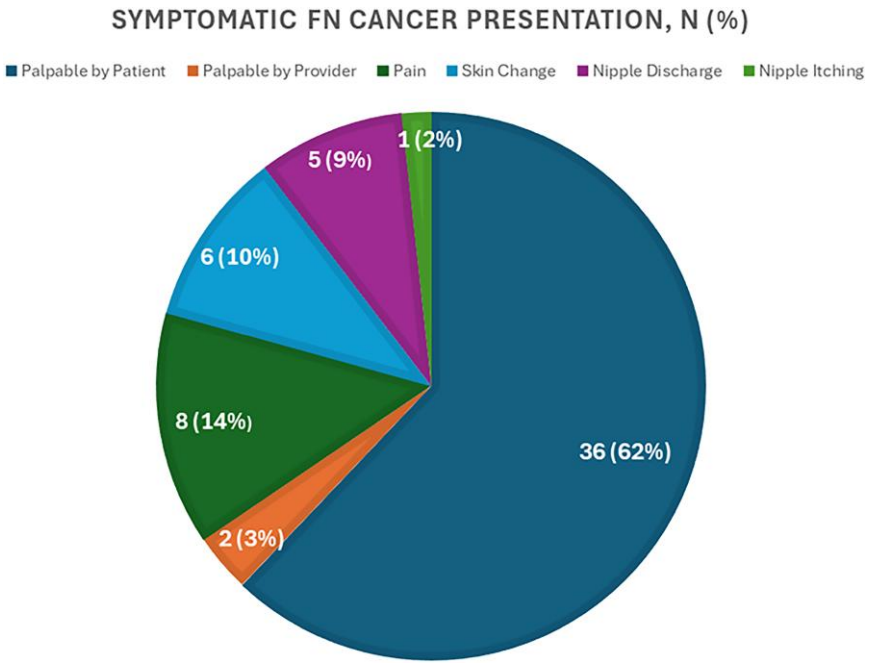


Figure 3. Presentation of symptomatic false-negative (FN) cancers by specific symptom(s) (*n* = 53 patients). Some FN cancers had more than 1 presenting symptom.

study by Durand et al and from 1.1 to 0.8 per 1000 examinations in the study by Bahl et al.[5,6] Given the differences in incremental CDR by SS modality, it is likely that the effect on the FNR would vary based on the type of SS offered, which is an additional factor to consider in the development of benchmarks. The benchmark as designed largely applies to symptomatic FNs. Practices with larger volumes of SS should consider subanalysis of FNs to understand the impact of asymptomatic cases detected on SS. Method-of-detection tracking will allow greater understanding of the overall effect of SS and perhaps whether a lower symptomatic FN benchmark is achievable. As SS volumes increase, additional multi-institutional analysis would be beneficial for understanding differences in symptomatic IC and asymptomatic FN and the corresponding effect on the FNR.

While prior studies have evaluated tumor characteristics of FN, relatively few studies have evaluated the impact of SS on the FNR and, to our knowledge, no prior studies have characterized the differences in patient demographics, risk status, or MOD among these unique categories of FN. A systematic review of the effect of SS on IC rates in mammography screening by Zeng et al demonstrated overall improved CDR with addition of SS with an associated modest reduction in symptomatic IC.[22] The authors highlighted that MRI had the most profound increase in CDR but also resulted in increased recall, biopsy, and cost, which are known limitations of MRI. Bahl

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Table 4. Utilization of SS MRI According to 2018 and 2023 American College of Radiology Recommendations by Patient Risk Category

		Number of patients who underwent SS MRI according to guideline year	
Symptomatic list of risks, *N* (% of column)		2018	2023
Traditional high-risk criteria	14 (48.3%)	1	1
PH cancer <50	6 (20.7%)	0	0
PH cancer ≥50 + dense tissue	4 (13.8%)	0	0
Dense tissue only	5 (17.2%)	NA	0
Total	29	1	1
Percentage of total who underwent recommended SS MRI by guideline year		4%	3%
Asymptomatic list of risks, *N* (% of column)		2018	2023
Traditional high-risk criteria	38 (62.3%)	31	31
PH cancer <50	5 (8.2%)	4	4
PH cancer ≥50 + dense tissue	1 (1.6%)	0	0
Dense tissue only	17 (27.9%)	NA	0
Total	61	35	35
Percentage of total who underwent recommended SS MRI by guideline year	-	80%	57%

Abbreviations: NA, not applicable; PH, personal history; SS, supplemental screening.

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et al studied the tumor characteristics of IC detected with DBT compared with digital mammography and, while they found no significant differences in CDR or FNR, the authors determined that asymptomatic FN cancers detected by MRI elevated the FNR from 0.8 to 1.1 per 1000, identical to the FNR in our study. The authors also found that MRI-detected asymptomatic FN cancers accounted for 25% of screening mammography FN cases, similar to our 31%, and had favorable tumor characteristics when compared with symptomatic IC.[5]

Our results demonstrated that most (74% [39/53]) patients with symptomatic IC were at higher-than-average risk for breast cancer, including 38% (20/53) having a PH of breast malignancy and 25% (13/53) with a TC score ≥20%. When analyzing this cohort further, we found that patients diagnosed with breast cancer after the age of 50 with any breast density were more likely to have a symptomatic IC. Other recent studies have demonstrated a similar increased risk of IC in survivors of breast cancer.[23,24] This is in keeping with the current ACR guidance for women with PH of breast cancer, regardless of age at diagnosis or breast density, to strongly consider SS with MRI, especially if other risk factors are present. However, even with full optimization of current guidelines, it is apparent that risk-based methods as we understand them remain insufficient. Improvements in risk assessment are needed to individualize SS recommendations to detect these aggressive cancers in vulnerable patients who are not otherwise categorized as higher-than-average risk under current criteria.

Previous studies, including an analysis of BCSC registry data by Nelson et al, have demonstrated that women with dense breast tissue are more likely to have an FN mammogram.[6,25] Our results confirm this, with 56% (67/119) of FN cancers occurring in women with heterogeneously or extremely dense tissue, 39% (46/119) and 18% (21/119) respectively. This highlights an important cohort of patients with FNs who could benefit from updated ACR guidelines and appropriateness criteria to utilize SS MRI: patients with dense tissue and no other known risk factors.[11,15] In our study, no patients with dense tissue as their only risk factor utilized MRI for SS. This is likely multifactorial, and while the updated recommendations for SS MRI in this category were released at the end of the study period, other factors, such as insurance coverage, likely contribute as well. Nearly one-third of patients with asymptomatic FN underwent SS US (in accordance with 2018 guidelines), and similar utilization may be observed in other programs, even if SS MRI is available, although it may vary depending on patient and provider familiarity with newer guidelines, local payer coverage, and current state laws regarding dense breast tissue SS.

Although a it is a small sample size, our results indicated that symptomatic ICs have a higher percentage of minority patients than asymptomatic FNs. While this was not statistically significant, breast cancer racial disparities are known, and similar results were demonstrated by Durand et al, with underlying reasons thought to be related to access, awareness, and/or choice of SS options.[6] Other social determinants of health have been shown to impact breast cancer outcomes, although our study was not designed to investigate these factors.

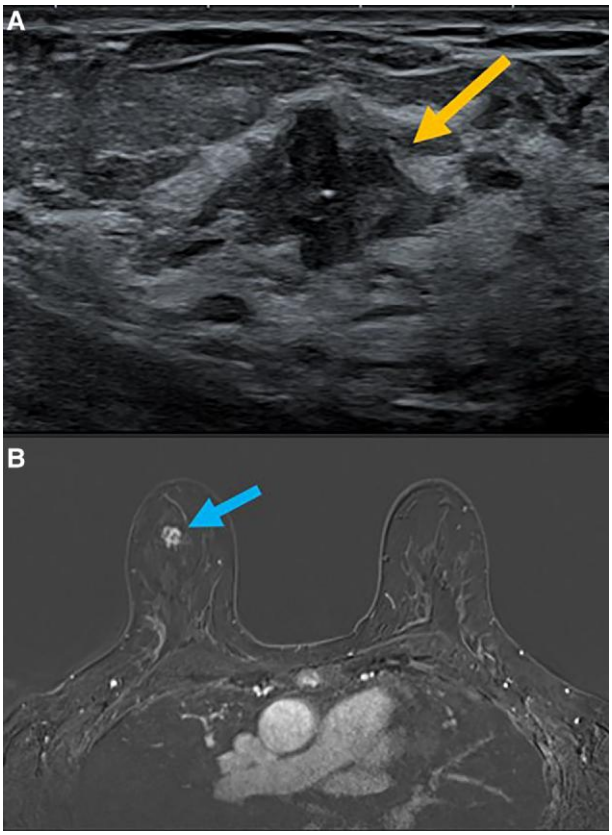


Figure 4. Forty-nine-year-old patient with a Tyrer-Cuzick >20% due to family history who was undergoing annual mammographic and MRI screening. Supplemental screening MRI 11 months prior and screening mammogram 1 week prior were negative. The patient presented with bilateral palpable lumps. Targeted right breast US demonstrates an irregular, hypoechoic mass containing calcifications at 11 o'clock (**A**; arrow). The palpable finding in the left breast was a benign cyst (not shown). US-guided biopsy yielded grade 2 invasive ductal carcinoma with intermediate-grade ductal carcinoma in situ. Retrospective analysis in this case demonstrated no suspicious findings on prior mammogram or MRI. Diagnostic MRI after the cancer diagnosis demonstrated the irregular, enhancing mass in the right breast and no additional suspicious findings (**B**; arrow). This patient underwent bilateral skin-sparing mastectomies with a tumor size of 1.6 cm and negative sentinel lymph node biopsy.

Our study results demonstrated that symptomatic IC are less favorable, with larger size at diagnosis and higher grade, than asymptomatic FN. Patients with symptomatic IC were also more likely to have morphologically abnormal lymph nodes at the time of diagnosis. These findings align with multiple previous studies evaluating all types of IC, including a review by Houssami and Hunter of 41 articles that found that ICs were consistently reported to be larger, higher grade, have less favorable biomarker profiles, and higher frequency of nodal metastases than screening-detected cancers.[2] Previous studies have also shown SS-detected cancers to be smaller

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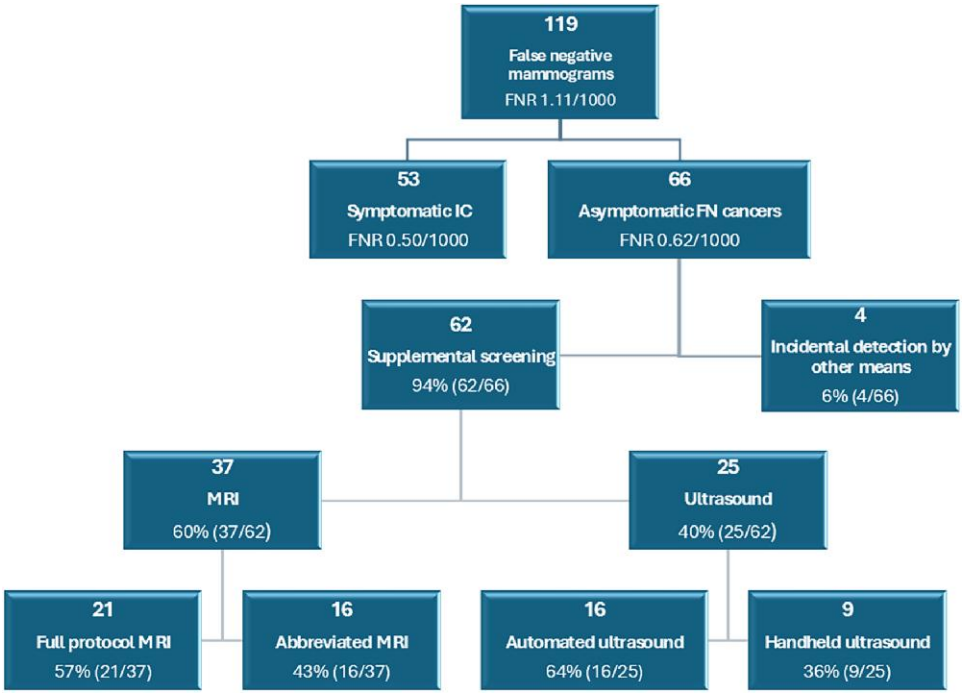


Figure 5. Breakdown of the study population with method of detection for asymptomatic FN cancers. Abbreviations: FN, false-negative; FNR, false-negative rate; IC, interval cancer.

and lower grade than those detected by screening mammography.[5,22] However, overall, few studies distinguish between symptomatic IC and asymptomatic FN, and even fewer characterize the differences in biological characteristics. Additional research will be necessary to confirm these findings and validate the role of SS in reducing morbidity and mortality from breast cancer by reducing symptomatic IC.

There are several limitations in this study. First, it was performed at a single academic institution with an established high-risk breast cancer prevention clinic, which may limit generalizability. Second, patient risk calculations were based on patient-provided information available to the radiologist at the time of interpretation rather than a formal risk assessment by a clinician. However, this likely reflects the level of data that can be obtained by most breast imaging facilities offering screening mammography services. Third, our practice began offering CEM toward the end of the study period, with no FN detected by CEM in our study population. The current FNR at our institution may be impacted even further by the addition of CEM, and programs offering contrast-based screening with MRI and CEM may experience higher rates of asymptomatic FNs given the higher reported sensitivities of these examinations. Fourth, it has been shown that approximately 20% to 25% of FN are cancers that were missed by the interpreting radiologist on the index screening mammogram.[2] Our study did not aim to retrospectively reinterpret screening mammograms, and while the internal

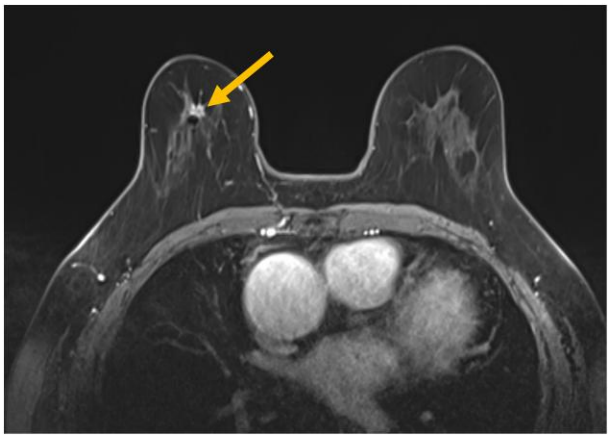


Figure 6. Fifty-eight-year-old patient with a personal history of breast malignancy at age 55 and heterogeneously dense breast tissue who underwent negative screening mammogram and handheld US (HHUS). At the recommendation of her oncologist, she switched supplemental screening modalities from HHUS to abbreviated MRI (AB MRI), and an AB MRI performed 6 months later demonstrated an 0.8-cm spiculated, enhancing mass at 12 o'clock, immediately adjacent to artifact from surgical clips (arrow). Targeted US was performed that revealed no correlate for the MRI finding, likely due to adjacent posttherapeutic findings. The patient then underwent MRI-guided biopsy and pathology yielded low-grade ductal carcinoma in situ, followed by breast conservation therapy. This patient was following both 2018 and 2023 American College of Radiology guidelines due to the recommendation from her provider.

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screening mammography CDR surpassed benchmarks, this may have provided further insight. Additionally, given that we did not perform a lesion subanalysis, we do not know whether the presenting symptom for the symptomatic IC corresponded to the subsequently diagnosed malignancy. While MOD currently captures data at the patient level, it would be interesting to understand the contribution of incidentally detected cancers within nonscreening MOD categories, and this could be a topic of future research. Lastly, we used the ACR 2018 and 2023 breast cancer screening guidelines to assess utilization of SS in our study population. While evidence supports current ACR recommendations for screening MRI, practical challenges related to cost and access remain. The 2023 guidelines specify that CEM can be considered as a potential SS method for patients with higher-than-average risk who have a contraindication to breast MRI. Likewise, patients with elevated risk who would qualify but cannot undergo MRI can consider SS with US, which has increased availability and lower implementation costs.[26]

Conclusion

Facilities offering robust SS programs for patients at higher-than-average risk may experience an elevated FNR above the historic benchmark of 0.8 per 1000 due to detection of asymptomatic FNs. With increasing utilization of SS, the distinction between symptomatic ICs and asymptomatic FN cancers helps diminish the confounding factor of SS on FNRs. This could provide a more meaningful medical audit and advanced categorical data for future research endeavors designed to understand and reduce ICs, which have been shown to have worse outcomes. Continued efforts to understand and improve individualized risk assessment are also needed to further tailor recommendations for each type of SS examination. Ongoing multi-institutional analysis of FNs will be helpful for understanding the impact on screening mammography benchmarks and allowing continued analysis of the characteristics of symptomatic ICs and asymptomatic FN cancers.

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Conflict of interest statement

None declared.

Author contributions

Ashley I. Huppe (Conceptualization, Data curation, Methodology, Project administration, Writing—original draft, Writing—review & editing), Onalisa Winblad (Conceptualization, Methodology, Supervision, Writing—review & editing), Jessica Peterson (Methodology, Writing—review & editing), Camron Smith (Methodology, Writing—review & editing), Molly Hill (Methodology, Writing—review & editing), Lauren Clark (Dara curation, Methodology), Carissa Walter (Data curation, Methodology), Allison Aripoli (Conceptualization, Data curation, Methodology, Project administration, Writing—original draft, Writing—review & editing)

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